# Probability Theory
# STA 1303

Rajarata University of Sri Lanka

# Student Information Plan

Course Title: Probability Theory

Course Code: STA 1303

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Tutor:

## Course Outline

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|---|---|
| Chapter 1 | Multivariate Probability Distribution |
| Chapter 2 | Functions of Random Variables |
| Chapter 3 | Special Sampling Distributions |
| Chapter 4 | Concepts of Stochastic Convergence and Central Limit Theorem |

# Multivariate Probability Distributions

# 1

# Introduction

Univariate

Bivariate

Multivariate

The intersection of two or more events is frequently of interest to an experimenter.

For example,

A gambler playing blackjack is interested in the event of drawing both an ace and a face card from a 52 -card deck.

A biologist, observing the number of animals surviving in a litter, is concerned about the intersection of these events:
A: The litter contains n animals.
B: y animals survive.

Suppose that $Y_1, Y_2, \ldots, Y_n$ denote the outcomes of n successive trials of an experiment.

For example, this sequence could represent the weights of n people or the measurements of n physical characteristics for a single person.

A specific set of outcomes, or sample measurements, may be expressed in terms of the intersection of the n events

$(Y_1 = y_1), (Y_2 = y_2), \ldots, (Y_n = y_n)$,

which we will denote as $(Y_1 = y_1, Y_2 = y_2, \ldots, Y_n = y_n)$, or, more compactly, as $(y_1, y_2, \ldots, y_n)$.

Calculation of the probability of this intersection is essential in making inferences about the population from which the sample was drawn and is a **major reason for studying multivariate probability distributions.**

# Multivariate Probability Distribution

Bivariate and Multivariate Probability Distribution

## Bivariate and Multivariate Probability Distribution

Many random variables can be defined over the same sample space.

For example, **consider the experiment of tossing a pair of dice.**

The sample space contains 36 sample points, corresponding to the mn = (6)(6) = 36 ways in which numbers may appear on the faces of the dice.

Any one of the following random variables could be defined over the sample space and might be of interest to the experimenter:

Y1: The number of dots appearing on die 1.

Y2 : The number of dots appearing on die 2 .

Y3: The sum of the number of dots on the dice.

Y4: The product of the number of dots appearing on the dice.

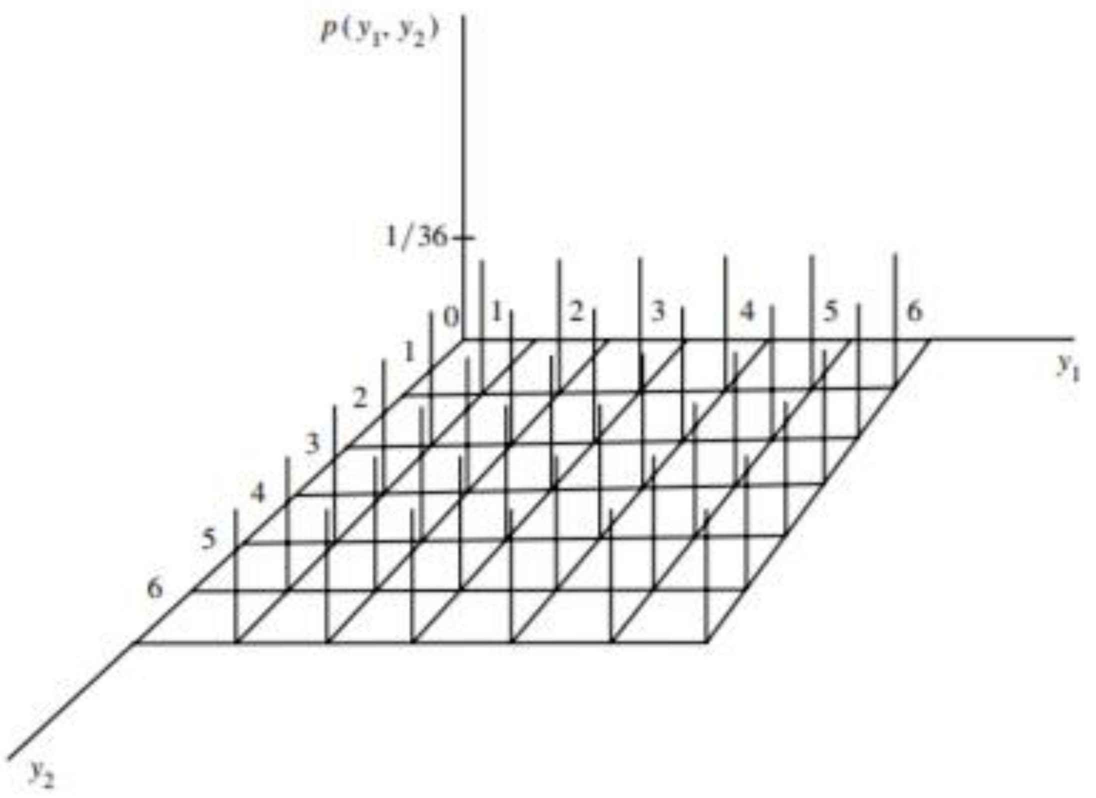
The 36 sample points associated with the experiment are equiprobable and correspond to the 36 numerical events $(y_1, y_2)$. Thus, throwing a pair of 1s is the simple event (1, 1).

Throwing a 2 on die 1 and a 3 on die 2 is the simple event (2 , 3). Because all pairs $(y_1, y_2)$ occur with the same relative frequency, we assign probability 1/36 to each sample point.

For this simple example, the intersection (y1, y2) contains at most one sample point.

Hence, the bivariate probability function is

$$\mathbf{p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2) = 1/36,\ y_1 = 1, 2, \ldots, 6,\ y_2 = 1, 2, \ldots, 6.}$$




Bivariate probability function;
$y_1$ = number of dots on die1,
$y_2$ = number of dots on die 2

## DEFINITION 1

Let $Y_1$ and $Y_2$ be discrete random variables.
The joint (or bivariate) probability function for $Y_1$ and $Y_2$ is given by

$$p(y_1, y_2) = P(Y_1 = y_1, Y_2 = y_2), -\infty < y_1 < \infty, -\infty < y_2 < \infty.$$

- ➔ In the bivariate case,
  The joint probability function $p(y_1, y_2)$ assigns nonzero probabilities to only a finite or countable number of pairs of values $(y_1, y_2)$.

- ➔ Further, the nonzero probabilities must sum to 1.

## THEOREM 1

If $Y_1$ and $Y_2$ are discrete random variables with joint probability function $p(y_1, y_2)$, then

1. $p(y_1, y_2) \geq 0$ for all $y_1, y_2$ .
2. $\Sigma_{y1,\, y2}\, p(y_1, y_2) = 1$, where the sum is over all values $(y_1, y_2)$ that are assigned nonzero probabilities.

- As in the **univariate discrete case**,
  the **joint probability function for discrete random variables** is sometimes called the **joint probability mass function** because it specifies the probability (mass) associated with each of the possible pairs of values for the random variables.

- Once the joint probability function has been determined for discrete random variables $Y_1$ and $Y_2$ , joint probabilities involving $Y_1$ and $Y_2$ is
  For the **die-tossing experiment**,

$$P(2 \le Y_1 \le 3,\ 1 \le Y_2 \le 2) \text{ is } P(2 \le Y_1 \le 3,\ 1 \le Y_2 \le 2)$$
$$= p(2\ ,\ 1) + p(2\ ,\ 2) + p(3,\ 1) + p(3,\ 2) = 4/36 = 1/9.$$

## Example 1

A local supermarket has three checkout counters. Two customers arrive at the counters at different times when the counters are serving no other customers. Each customer chooses a counter at random, independently of the other. Let $Y_1$ denote the number of customers who choose counter 1 and $Y_2$ , the number who select counter 2 . Find the joint probability function of $Y_1$ and $Y_2$.

## DEFINITION 2

For any random variables $Y_1$ and $Y_2$ , the joint (bivariate) distribution function $F(y_1, y_2)$ is

$$F(y_1, y_2) = P(Y_1 \le y_1, Y_2 \le y_2), \quad -\infty < y_1 < \infty, \quad -\infty < y_2 < \infty.$$

For two discrete variables $Y_1$ and $Y_2$ , $F(y_1, y_2)$ is given by

$$F(y_1, y_2) = \sum_{t_1 \le y_1} \sum_{t_2 \le y_2} p(t_1, t_2).$$

For the die-tossıng experıment,

$F(2 , 3) = P(Y_1 \le 2 , Y_2 \le 3) = p(1, 1) + p(1, 2) + p(1, 3) + p(2 , 1) + p(2 , 2) + p(2 , 3).$

Because $p(y_1, y_2) = 1/36$ for all pairs of values of $y_1$ and $y_2$ under consideration,

$$F(2 , 3) = 6/36 = 1/6.$$

## Example 2

Consider the random variables $Y_1$ and $Y_2$ of Example 1. Find $F(-1, 2)$, $F(1.5, 2)$, and $F(5, 7)$.

Two random variables are said to be jointly continuous if their joint distribution function $F(y_1, y_2)$ is continuous in both arguments.

# DEFINITION 3

Let $Y_1$ and $Y_2$ be continuous random variables with joint distribution function $F(y_1, y_2)$. If there exists a nonnegative function $f(y_1, y_2)$, such that

$$F(y_1, y_2) = \int_{-\infty}^{y_1} \int_{-\infty}^{y_2} f(t_1, t_2)\, dt_2\, dt_1,$$

for all $-\infty < y_1 < \infty$, $-\infty < y_2 < \infty$, then $Y_1$ and $Y_2$ are said to be jointly continuous random variables. The function $f(y_1, y_2)$ is called the joint probability density function.

Bivariate cumulative distribution functions satisfy a set of properties similar to those specified for univariate cumulative distribution functions.

## THEOREM 2

If $Y_1$ and $Y_2$ are random variables with joint distribution function $F(y_1, y_2)$, then

1. $F(-\infty, -\infty) = F(-\infty, y_2) = F(y_1, -\infty) = 0$.
2. $F(\infty,\infty) = 1$.
3. If $y_1^* \geq y_1$ and $y_2^* \geq y_2$, then

$$F(y_1^*, y_2^*) - F(y_1^*, y_2) - F(y_1, y_2^*) + F(y_1, y_2) \geq 0.$$

# THEOREM 2

If $Y_1$ and $Y_2$ are jointly continuous random variables with a joint density function given by $f\,(y_1, y_2)$, then

- $f\,(y_1, y_2) \ \geq\ 0$ for all $y_1, y_2$ .
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f\,(y_1, y_2)\, dy_1\, dy_2 = \ 1.$

## Example 3

Suppose that a radioactive particle is randomly located in a square with sides of unit length. That is, if two regions within the unit square and of equal area are considered, the particle is equally likely to be in either region. Let $Y_1$ and $Y_2$ denote the coordinates of the particle's location. A reasonable model for the relative frequency histogram for $Y_1$ and $Y_2$ is the bivariate analogue of the univariate uniform density function:

$$f(y_1, y_2) = \begin{cases} 1, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

a) Sketch the probability density surface.
b) Find $F(.2\,,.4)$.
c) Find $P(.1 \le Y_1 \le .3, 0 \le Y_2 \le .5)$.

# Q1

Contracts for two construction jobs are randomly assigned to one or more of three firms, A, B, and C. Let $Y_1$ denote the number of contracts assigned to firm A and $Y_2$ the number of contracts assigned to firm B. Recall that each firm can receive 0, 1, or 2 contracts.

a) Find the joint probability function for $Y_1$ and $Y_2$ .

b) Find $F(1, 0)$.

# Q2

Three balanced coins are tossed independently. One of the variables of interest is $Y_1$, the number of heads. Let $Y_2$ denote the amount of money won on a side bet in the following manner. If the first head occurs on the first toss, you win \$1. If the first head occurs on toss 2 or on toss 3 you win \$2 or \$3, respectively. If no heads appear, you lose \$1 (that is, win −\$1).

a) Find the joint probability function for $Y_1$ and $Y_2$ .

b) What is the probability that fewer than three heads will occur and you will win \$1 or less? [That is, find $F(2, 1)$.]

# Q3

Let $Y_1$ and $Y_2$ have joint density function

$$f(y_1, y_2) = \begin{cases} e^{-(y_1+y_2)}, & y_1 > 0, y_2 > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

a) What is $P(Y_1 < 1, Y_2 > 5)$ ?

b) What is $P(Y_1 + Y_2 < 3)$ ?

# Q4

Let $Y_1$ and $Y_2$ have the joint probability density function given by

$$f(y_1, y_2) = \begin{cases} k y_1 y_2, & 0 \le y_1 \le 1, 0 \le y_2 \le 1, \\ 0, & \text{elsewhere.} \end{cases}$$

a) Find the value of k that makes this a probability density function.

b) Find the joint distribution function for $Y_1$ and $Y_2$ .

c) Find $P(Y_1 \le 1/2 , Y_2 \le 3/4)$.

# Q5

Let $(Y_1, Y_2)$ denote the coordinates of a point chosen at random inside a unit circle whose center is at the origin. That is, $Y_1$ and $Y_2$ have a joint density function given by

$$f(y_1, y_2) = \begin{cases} \dfrac{1}{\pi}, & y_1^2 + y_2^2 \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$$

Find $P(Y_1 \leq Y_2)$.